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TNF Future Light







Standard Operation

PT. YONGJIN JAVASUKA

1. Purpose

The purpose of this procedure is to outline the standard operation required to achieve Future Light branding for "The North Face" outerwear products. This is the minimum requirement and any non-adherence must be discussed and approved on a case-by-case basis with TNF Development/Quality Assurance team.

2. Scope

The standards apply to any style or production looking to include Future Light as part of the branding of the materiality determination of the product.

3. Quality Aspects of raw material

3.1. Evaluation of raw material quality

3.1.1 The Future light fabric mill needs to conduct the 100 % fabric inspection, including both face and back side following by 4-point system. And highlight all defect points for garment manufacturing management.

3.1.2 The Future Light fabric mill needs to ensure the maximum of fabric heat shrinkage is 4%.

3.1.3 Fabric mill must submit the passed 3rd party laboratory test report for the first shipment and in- house test report for each dye lot. Testing criteria and method must comply with Future Light MPS.

3.2 Storage

3-2-1 Grouping: Fabric mill must separate all rolls into 3 groups follow by below role:

Group A : Below 75 yards

Group B : 75 – 100 yards

Group C : 100 – 125 yards

Mill should store fabric per group with supportive Styrofoam, each layer of fabric should have 3 supportive Styrofoam place in aliquots position before adding additional layer, also stacking fabric over then 6 layers is prohibited

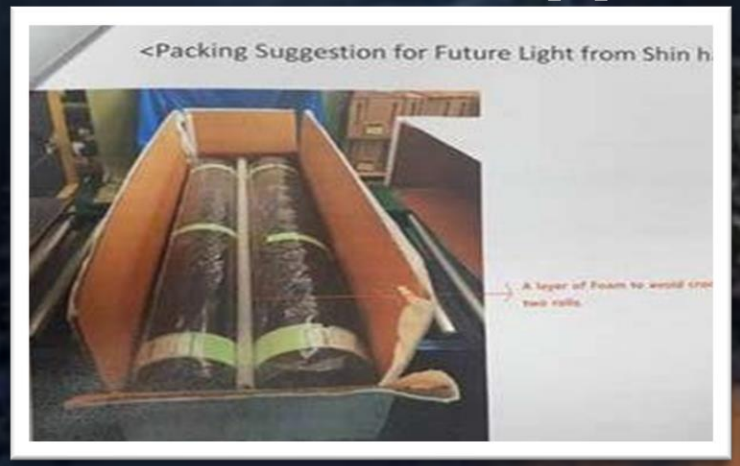




3-3. Packing

3.3.1. Packing method A, pack with paper carton.

3.3.1.1 All Future Light should pack rolls into a carton per group, mixed group rolls in one box are prohibited. Mill should insert foam paper between rolls to avoid uneven force increase pressure points





3.3.2 Packing method B, pack without carton box

3.3.2.1 Mill ship FL fabric with supportive Styrofoam. Each layer should have 3 supportive Styrofoam place in aliquots position also 6 layers are the maximum height. Please referee below photo as standard packing method B.





4. VALUATION OF RAW MATERIAL QUALITY

4.1 Inspection

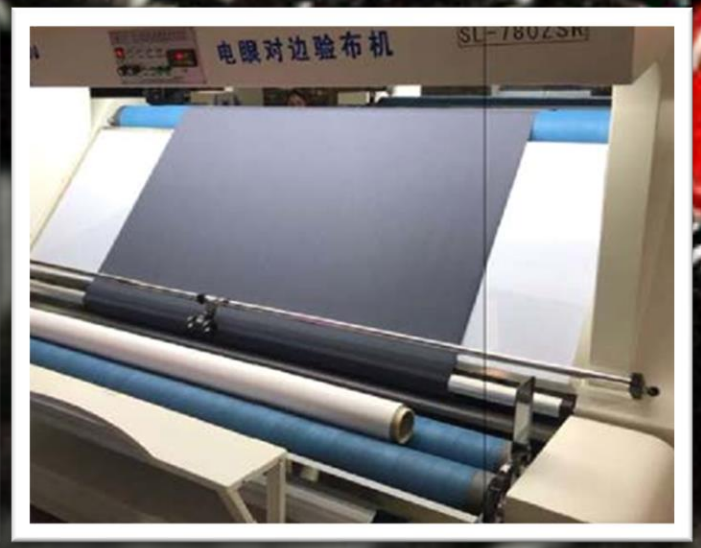
Fabric construction quality shall be evaluated utilizing a "Four Point" system for grading d

"FOUR POINT" PENALTY SCORING	
Points	**CRITERIA**
1 point	For defects up to 3" in any direction
2 points	For defects from 3" – 6" in any direction
3 points	For defects from 6" – 9" in any direction
4 points	For defects over 9" in any direction

4.1.1 Inspection Sampling Plan

10% of each dye lot is requested. All inspection reports must keep for 18 months for tracking





5. Material Storage

5.1. The Future Light fabric

5.1.1. The Future Light fabric shall store inside garment factory warehouse dry and clean avoid direct sunshine.

5.1.2. Future Light fabric shall store with standard rack to eliminate uneven pressure/ force flat hard board in both side of the rack and under the rack is requested or used pipe to eliminate fabric crease mark and hard pressure

5.1.3. Stacking Future Light fabric over then 3 layers is prohibited.





5.2. The Seam seal Tape/ Adhesives

5.2.1. All Seam seal Tape and Adhesives must be stored inside conditional room (Temperature < 25 degrees, RH < 65%)

5.2.2. List and put separately of seam tape expire date (as useful life of Bemis Adhesives) can't be used more than 2 years (see supplier e-mail)







Expired more 2 years



PT. YONGJIN JAVASUKA

6. Pre-Production

6.1. Mock-up testing: The purpose of mock-up testing requirement is to ensure the parameter / recipe is suitable for mass production. All embellishments must conduct required testing before PP meeting and all results must be presented during PP meeting.

6.1.1. Heat transfer print test requirement

Heat transfer print mock-up test requirement		Requirement
Washing test	VFOI-004 20 cycles wash (40 degree) and tumble dry low (wash + dry) *20 times	Class 4.0
Oven test	VFOI-007 70 degree * 24 hrs	Class 4.0

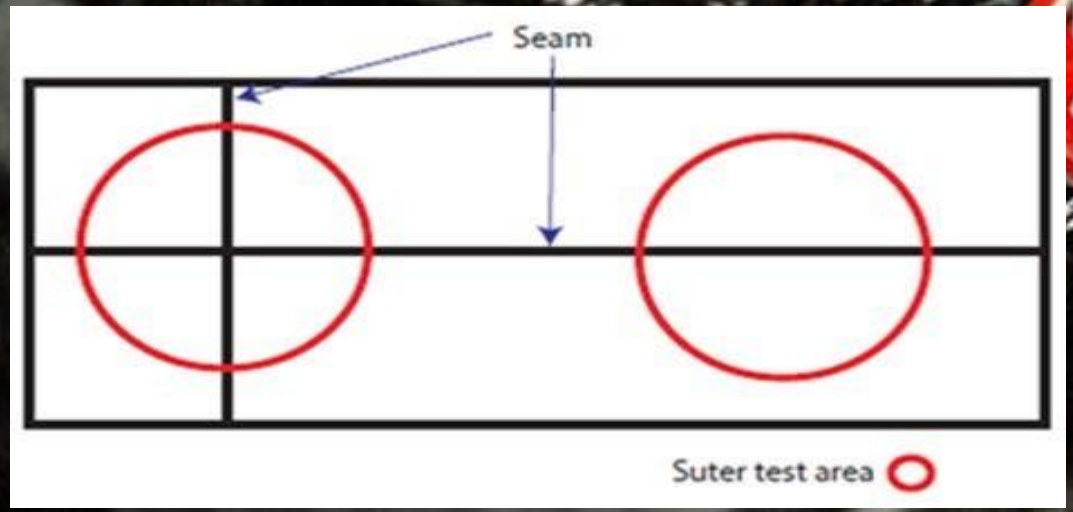
6.1.2. Screen print test requirement

Heat transfer print mock-up test requirement		Requirement
Washing test	VFOI-004 20 cycles wash (40 degree) and tumble dry low (wash + dry) *20 times	Class 4.0
Oven test	VFOI-007 70 degree * 24 hrs	Class 4.0
Crocking	AATCC 8	Dry - Class 4.0; Wet - Class 3.0

6.1.3. Seam sealing test requirement

Seam Sealing mock-up test requirement		Requirement
Washing test	VFOI-004 20 cycles wash (40 degree) and tumble dry low (wash + dry) *20 times	Class 4.0
Oven test	VFOI-007 70 degree * 24 hrs	Class 4.0
Hydrostatic test	**1.5 psi/ 1 mins** (before wash)	***AATCC 127, it states to report failures "disregarding water droplets that appear within approximately 3 mm of the specimen clamping ring".

Seam Sealing Mock up Requirement



6.1.4. Bonding test requirement

Bonding mock -up test requirement		Requirement
Washing test	VFOI-004 20cycleswash(40degree)andtumbledrylow(wash+dry) *20 times	Class 4.0
Over test	VFOI-007 70 degree * 24 hrs	Class 4.0
Pulling	Shear - before and after wash	30lbs
	Peel	3lbs/ inch
	Bonding zipper – before and after wash :	10 lbs for 10sec

6.2. Pre-production meeting

Pre-production meeting should be hold 5 days before cutting. Following samples and documents must be presented and reviewed during PP meeting by CFA or In house QA

- PP-sample + POC Sample
- PP-sample comments + POC Sample Comment
- Pre-production mock-up test result and specimens
- Fabric test reports provided by fabric mill
- Supplier recomendation (MDS or PDS) for snap, Bemis, H/T, I/L and seam sealing



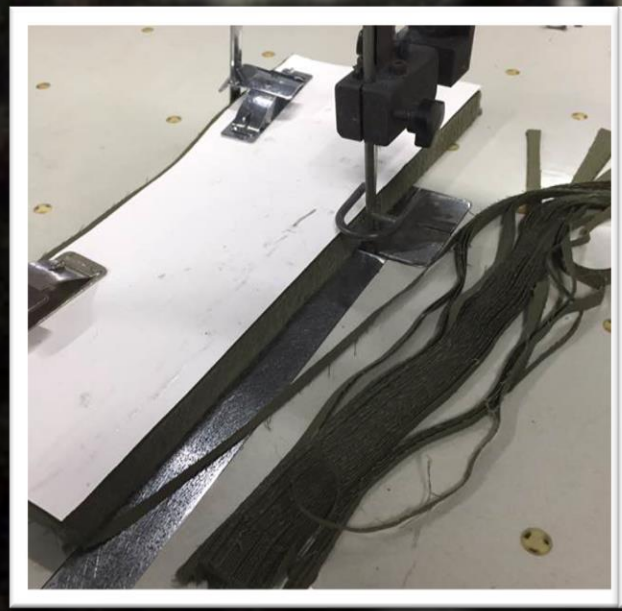
7. Cutting Section

7.1. Fabric relaxation: All Future Light fabric must relax based on stretch properties before cutting. 2 x 24 Hours and keep record the fabric

7.2. Cutting Process

7.2.1. Manual cut: any sharp or heavy weight objects are prohibited on cutting table. Metal clips must be covered with fleece fabric, sponge or any soft items.

7.2.2. Automation cut: the height of 1 batch or spread layer must lower than 2.5".







7-3 Bundling

Roll up panels in bundles with lamination faced inside. Bundles are not too tight (Any bundle mark caused by rope or tape).







8. Seam sealing

8.1. Mock Up test and Machine Calibration

8.1.1. Receive PDS (Product Data Sheet) or MDS (Material Data sheet) from fabric mill / supplier to determine the parameter of machine condition.

8.1.2. Set up seam sealing machine based on parameter received from fabric mills and seam tape supplier by setting the parameter of temperature, speed, and pressure.

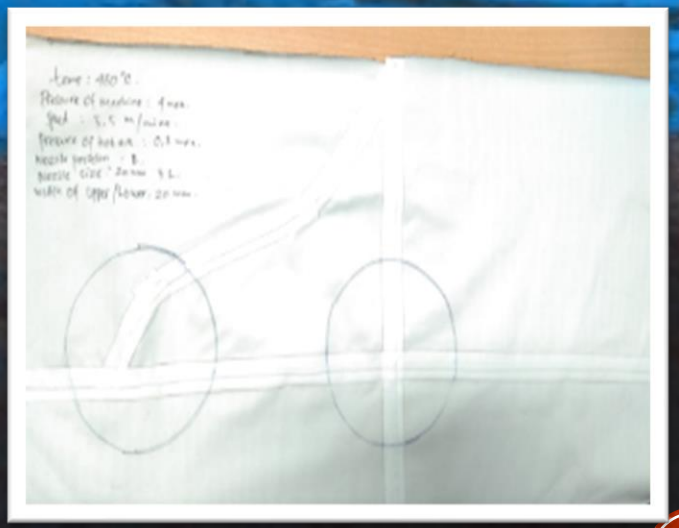
8.1.3. Make the mock up to test seam tape application as requirement for Future Light fabric for all color way. Mock up must be made2 set of each color before running bulk production.

8.1.4. Do water leakage (waterproof) test of the mock up before and after washing. Analyze the result of mock up washing test, then re-set up the machine by adjusting the parameter if there is issue and problem on the mock up. If "OK" mentioned the corrected machine set up on the mock up for production guideline

"





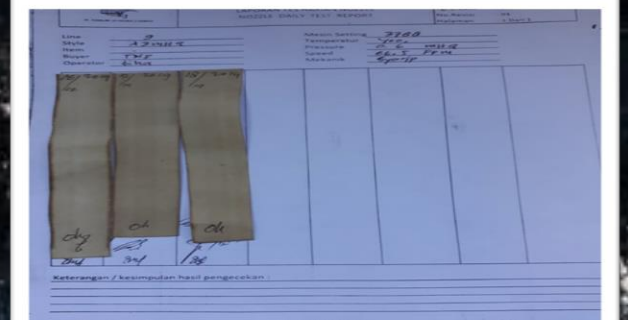


8.1.5 Set up the fixed parameter of machine for seam tape application if mock up test result is OK

.8.1.6. Nozzle heat testing: checking with temperature paper 2 times a day. (Before starting production in the morning and after lunch break)

8.1.7 Please Do not run seam sealing process for bulk production of machine setting still not OK, unless otherwise the machine is set up correctly.





8.1.8 Make sure the panel that will be sealed is clean, no threads end, no puckering and stitching quality is good.

8.1.9. Do not pull the fabric during seam sealing, and make sure the handling by operator is correct.

8.1.10 Seam tape must be in the center of seam, seam tape must be even and Seam allowance must be
evenly cut and tidy 2-3mm

8.1.11 Seam tape allowance must be 10mm, Overlap seam tape for joining is 10mm

8.1.12 All seam tape edge must be no lifting edge or peel off

8.1.13. Used Nozzle type as per seam tape quality (2 Layer or 3 Layer seam tape)

using different nozzle type to make even spreading of the heat so that sealing camelted to the fabric strongly

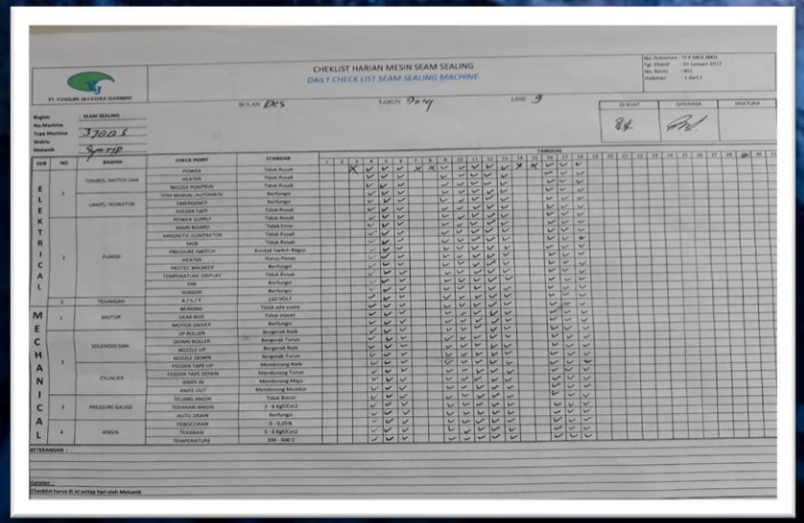


Nozzle 3 Layer



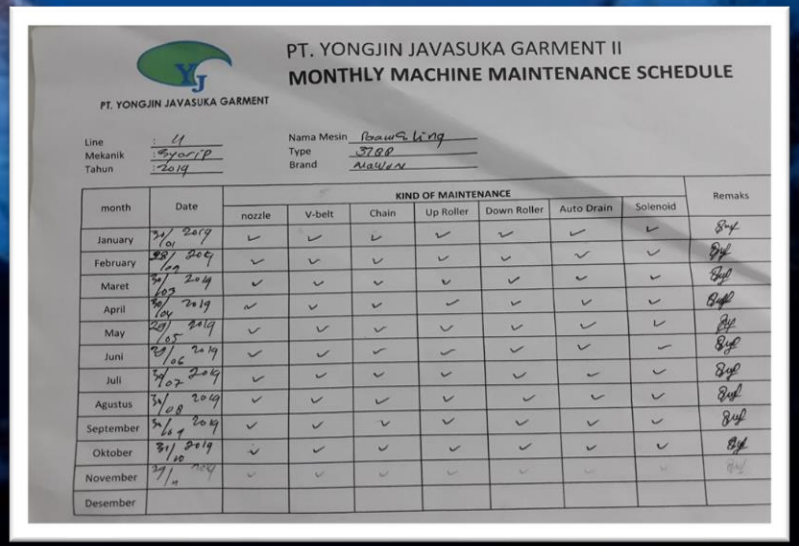
Nozzle 2 Layer

8.1.14. Daily, monthly and annually machine maintenance in place. All records must keep for 12 months for process audit and reference. Nozzle heat: Controlling heat air even by nozzle checking twice per day (before start production in the morning & after lunch time).



CHECKLIST HARIAN MESIN SEAM SEALING

DAILY CHECK LIST SEAM SEALING MACHINE



PT. YONGJIN JAVASUKA GARMENT II

MONTHLY MACHINE MAINTENANCE SCHEDULE

PT. YONGJIN JAVASUKA GARMENT

month	Date	KIND OF MAINTENANCE							Remaks
		nozzle	V-belt	Chain	Up Roller	Down Roller	Auto Drain	Solenoid	
January	31/01 2019	✓	✓	✓	✓	✓	✓	✓	
February	28/02 2019	✓	✓	✓	✓	✓	✓	✓	
Maret	31/03 2019	✓	✓	✓	✓	✓	✓	✓	
April	30/04 2019	✓	✓	✓	✓	✓	✓	✓	
May	29/05 2019	✓	✓	✓	✓	✓	✓	✓	
Juni	29/06 2019	✓	✓	✓	✓	✓	✓	✓	
Juli	31/07 2019	✓	✓	✓	✓	✓	✓	✓	
Agustus	31/08 2019	✓	✓	✓	✓	✓	✓	✓	
September	30/09 2019	✓	✓	✓	✓	✓	✓	✓	
Oktober	31/10 2019	✓	✓	✓	✓	✓	✓	✓	
November	31/11 2019								
Desember									

8.1.15. Seam Sealing Setting Future Light process for PPS S # A4R24 / A4NGT

DETAIL SETTING SEAMSEALING MACHINE FOR FL FABRIC

MATERIAL : LYON FOR FABRIC SUPP : SHINHAN
LV7Q FOR SEAMTAPE SUPP : SEALON

SEAMSEALING MACHINE CONDITION :

- TEMPERATURE 480° C
- PRESSURE 4 Mpa
- SPEED 3.5 Rpm
- NOZLE for 3 layer fabric

CROSSBAR CONDITION :

- TEMPERATURE 130° C
- PRESS 5 second

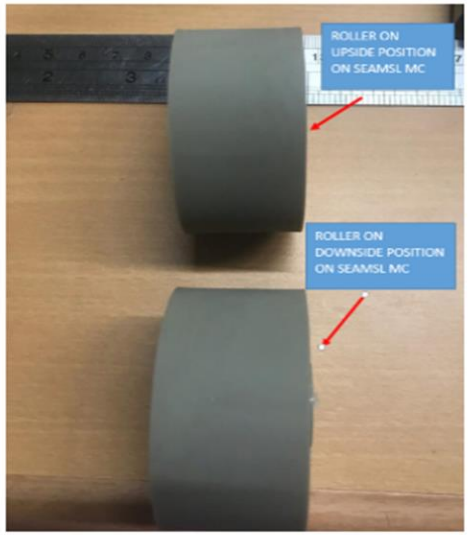
SEAMSEALING MACHINE



NOZLE WITH FOR 3 LAYER FABRIC, NEW VERSION



ROLLER SEAMSEALING MACHINE



SIDE SEAM UNCROSSBAR, SUTER TEST RESULT IS PASS





9. Pressing (Cross Bar)

9.1. Definition / Purpose

- Pressing (cross ba) is required to reinforce seam sealing (seam tape) of water proof garment to av avoid leakage seam and feel off seam tape or seam sealing application

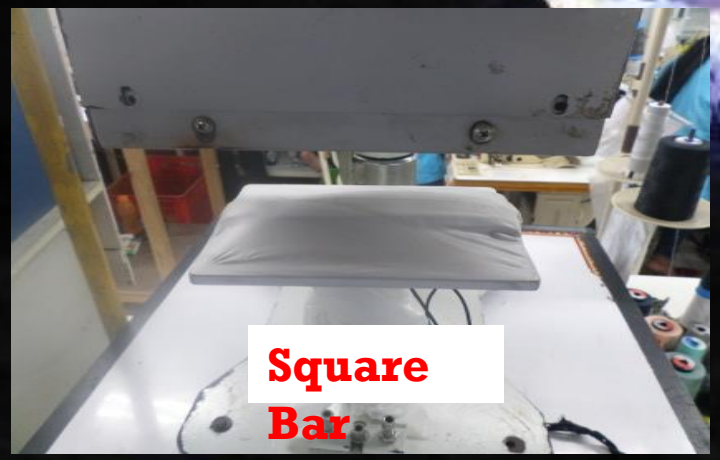
9.2. Setting machine

Set up machine press machine (cross bar) by correct parameter according to characteristic of fabric

9.3. Working Procedure

Use the pressing machine (cross bar) with suitable bar to press seam tape application on the garment.

1. Cross bar machine with square bar
2. Cross bar machine with rounded bar



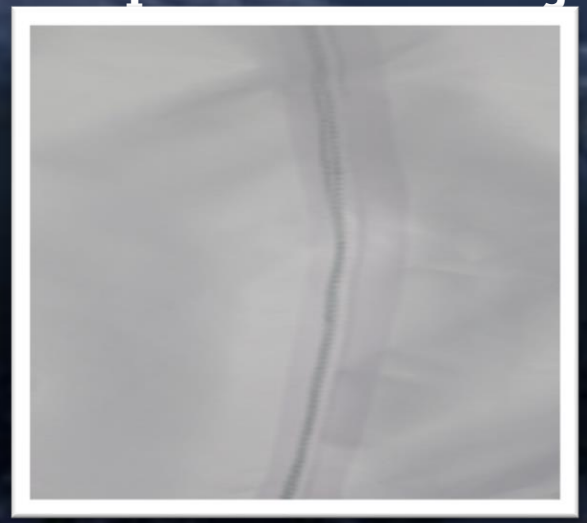
Square Bar

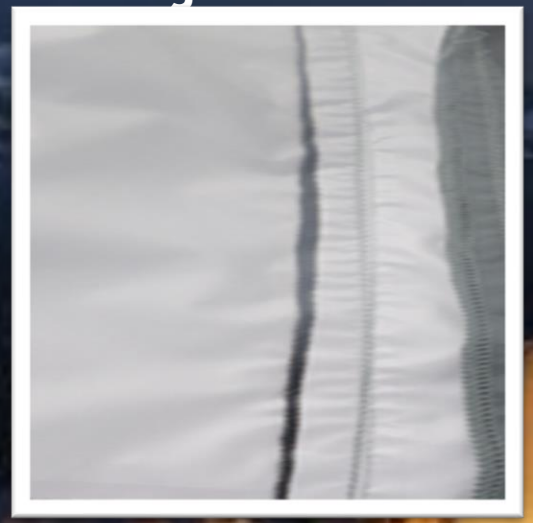


Rounded Bar

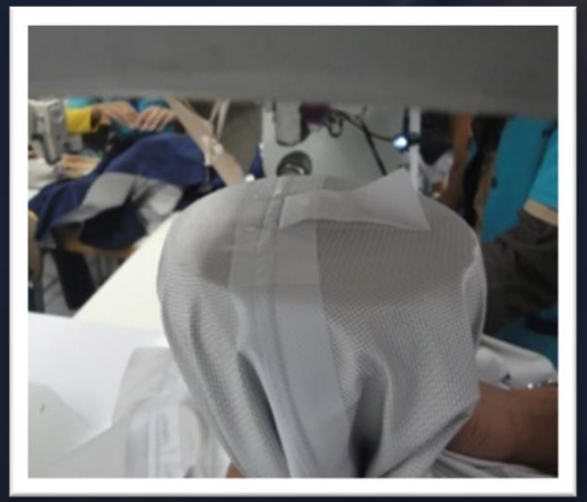
Type of coated fabric	Temperature (Degree)	Times (Second)
Normal coated fabric / PU coated fabric	120 – 150	3 – 5
2 layer coated fabric	150 – 180	3 – 5
3 layer coated fabric	180 – 220	3 – 5

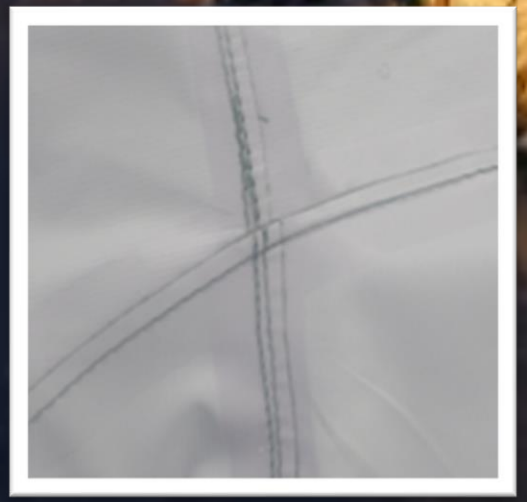
9.3.1 Use square bar for straight or single seam and center front (center front zipper)

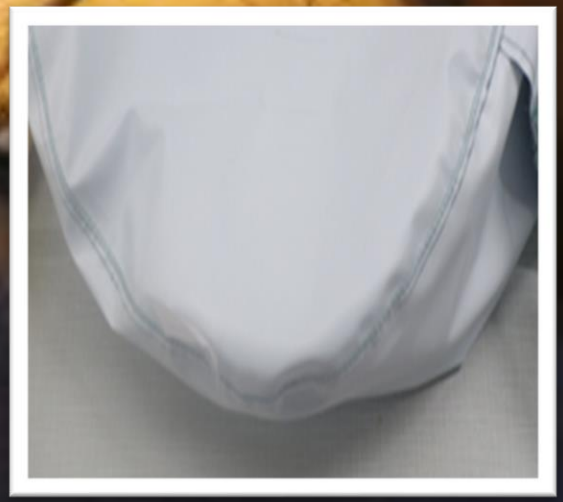




9.3.2 Use the rounded bar for tacking seam sealing, cross seam or T-cross seam, curve seam and hood, neck seam and armhole seam.





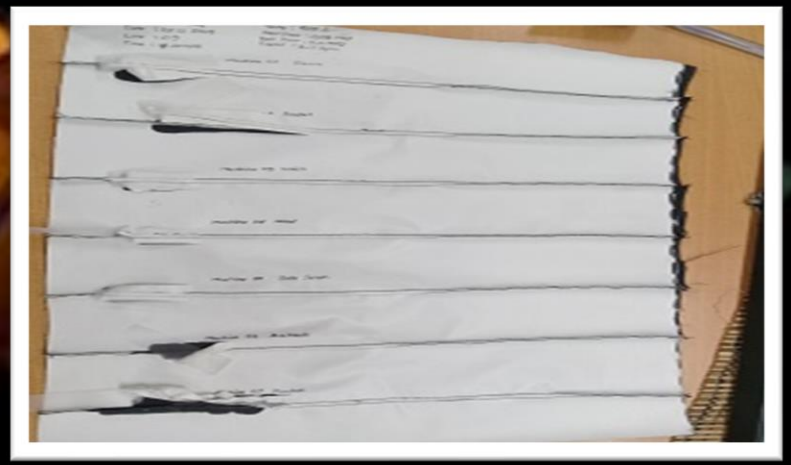


9.3.3 Fix the seam of the garment on the bar tidy, no pleated and crease.
Do pressing (Cross Bar) properly, (NOT RUSHING) in accordance with predetermined time.





9.3.3 Prepare seam sealing mock for each machine to ensure the seam tape strong enough to melted on the fabric
2 times a days (morning and after lunch)



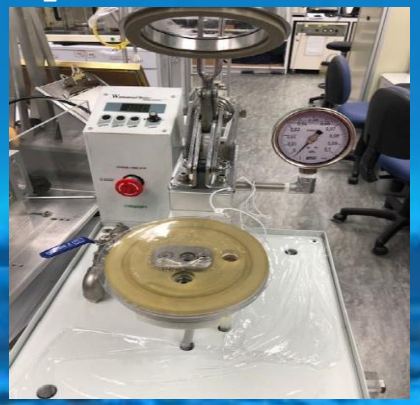


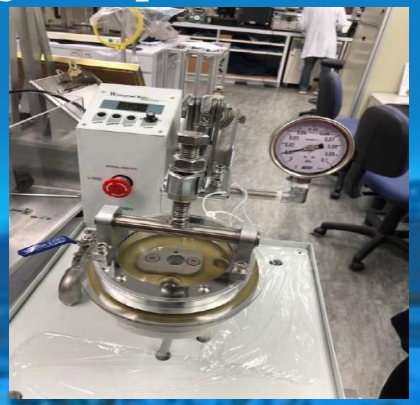
10. Future Light Water Proof Standard

10.1 Water Proof process

10.1.1. Set up waterproof test machine with correct parameter according to requirement standard.

Future Light : 1.5 PSI / 1 mnts
Other fabrication must refer to
product data sheet for each fabric supplier
Suter test machine should calibrated with
plastic sheet to ensure the pressure in
1.5 psi before testing.

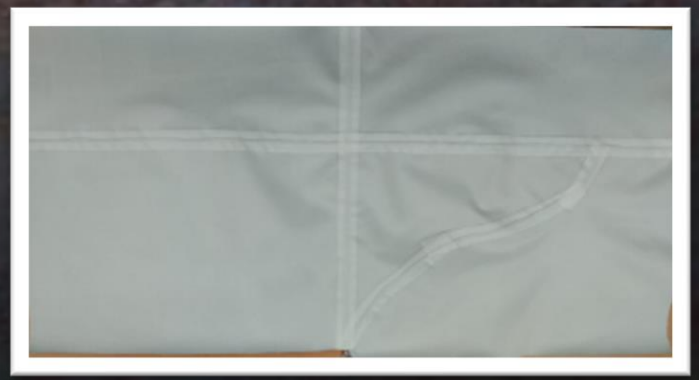




10.1.1. Future Light Fabric water leakage test can't do direct to the garment to avoid fabric Damage, mechanic /production have to prepare seam sealing mock up 2 time a day /machine (morning and after lunch)

For safety (Yongjin suggested) to test mock up 4 times a day (7.30 am, 10.30 am and 14.00 pm , 15.30 pm)





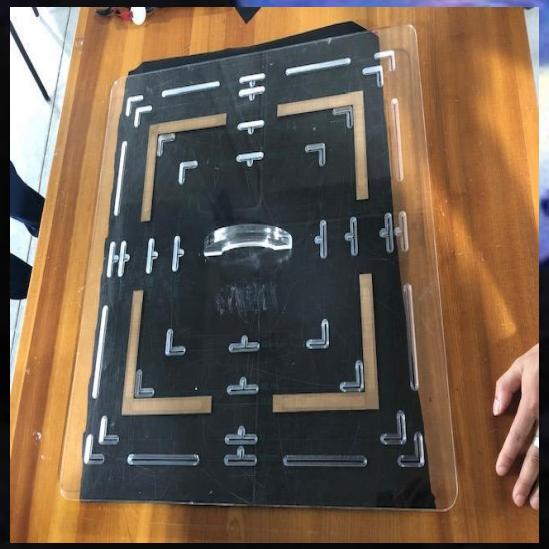
10.1.2 Do mock up shrinkage test before stand seam sealing process with following :

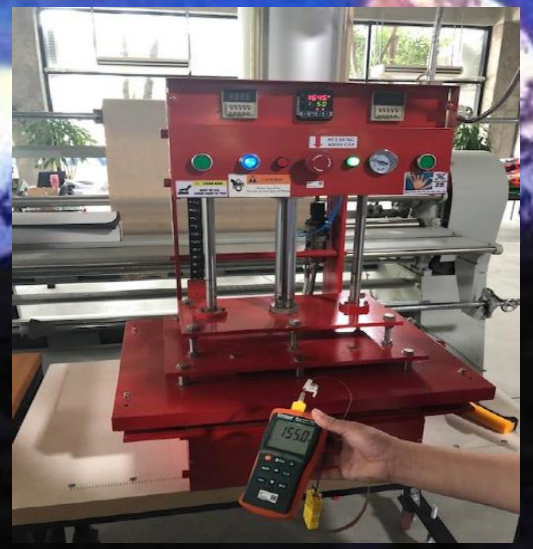
1. Relaxed/condition fabric for mock up on 4 hours before marking by 10 “ x 10”
2. Use template /pattern for marking 10” x 10 “ (as per AATCC standard)
3. Do pressing the mock up with below machine setting

   Glue line Temperature: 150 ˚C

   Hot time: 20 second

   Pressure: 0.25 mpa
4. Relaxing / condition mock up after pressing along 4 hours
5. Measure mock width and length and calculated before and after pressed if incase mock up shrinkage more 3% we must voice out to the supplier and if shrinl within tolerance +- 3% we have to voice out and adjust the pattern







10.1.3 Record daily test mock up result on the waterproof test report. 4 times /day

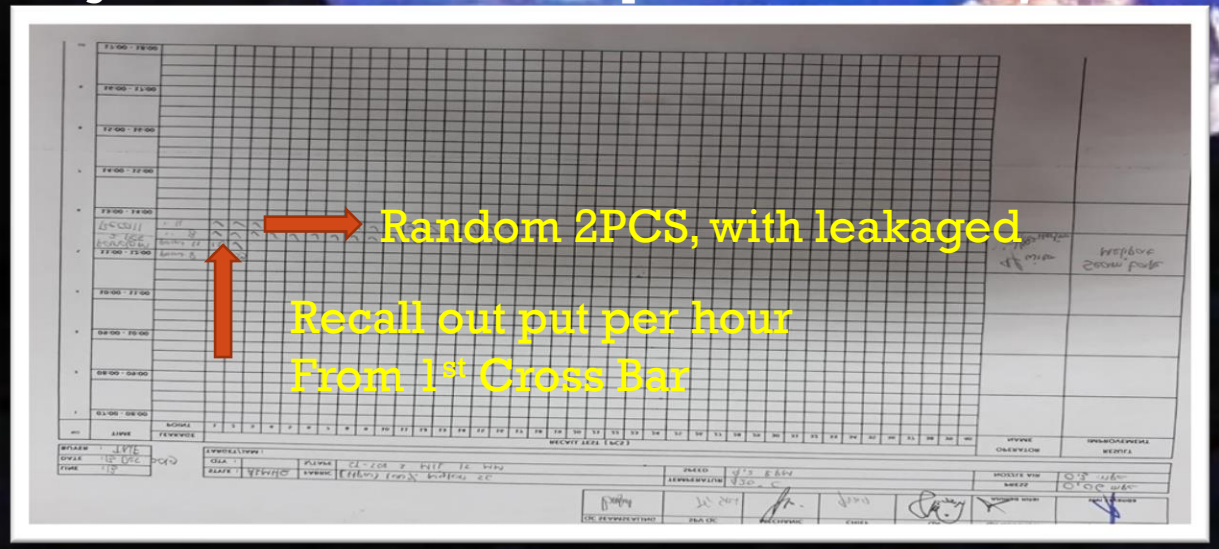
DAILY SUTER TESTING REPORT FOR FUTURELIGHT MOCK-UP - BEFORE & AFTER WASH

Date :

No.	Program/ Monthly Buy	Hours	Style#	Color #	Prod Line No#/	Seam Sealing No#	Heat Press No#	Mockup testing week	Mock-Up Testing Date	Suter Testing Standard	Suter Testing Mock-Ups (Before Wash) - 8:00AM			After Wash Moc-Ups (Washing For Mock-Ups: 8:00AM)	Suter Testing Mock-Ups (Before Wash) - 13:30PM			After Wash Mock-Ups (Washing For Mock-
											Testing Point No#1	Testing Point No#2	Remark - 8:00AM		Testing Point No#1	Testing Point No#2	Remark - 1:30PM	
1	TNFF20/ Dec Buy	07.30 AM	NF04R24	JK3	L12	1	1			1.5 PSI/ 1 MIN								
2																		
3																		
4																		
5		10.30 AM	NF04R24	JK3	L12	1	1			1.5 PSI/ 1 MIN								
6																		
7										1.5 PSI/ 1 MIN								
8		14.00 PM	NF04R24	JK3	L12	1	1											
9																		
10										1.5 PSI/ 1 MIN								
11																		
12		15.30 PM	NF04R24	JK3	L12	1	1											
13																		
14										1.5 PSI/ 1 MIN								
15										1.5 PSI/ 1 MIN								
16										1.5 PSI/ 1 MIN								

10.1.4 Immediately report to sewing manager or Mechanic if waterproof test is leakage.

10.1.5 If found leakage on seam or cross seam, garment must be recalled to be repaired and seam sealing machine must be re-set up and record in daily recall form report



PT. YONGJIN JAVASUKA

10.1.6 Recall Process

1. All garments after cross seam press must have color tie (Raffia) to trace the garments of particular one hour output. (identification as one batch output)

2. Each garments should have different tie color (raffia) per one hour output.

3. Aside from putting color tie on each garments, Cross seam press operator must also have counter (record) to quantify how many output have finished within 1 hour.

4. (Yongjin suggested for safety) to do Sutter test mock up 4 time / day (7.30 , 10.30 am and 14.00 pm , 15.30 pm)

5. If in case there's failed mock up water test at 10.30 am , operator need to conduct recalling all garments from output sewing 7.30 to 10.30 and if incase failed on 15.30 pm so garment output from 14.00 pm to 15.30 must be recalled .

6. if there's a failed water test, operator need to raised alert by calling the line supervisor/QC head and mechanic.

7. During in-line patrol by Factory QA manager, factory roving QC, VFQA, CFA need to review the actual Execution of Seam sealing operation & Sutter test if properly executed. They should sign the report showing what date/time they verified the process and put their remarks/findings.

11. Bonding Heat Press

11.1. Machine Calibration

11.1. Template actual temperature checking machine actual temperature every 2 hours per machine.

11.2 Mock-up testing





Bonding mock -up test requirement		Requirement
Washing test	VFOI-004 3 cycles wash (40 degree) and tumble dry low (wash + dry) *3 times	Class 4.0
Pulling	Shear - before and after wash	30lbs
Pulling	Peel	3lbs/ inch
	Bonding zipper – before and after wash:	10 lbs for 10sec

11.2. Requirement of Auxiliary tools

11.2.1. Using Acetone for removing stains from fabric surface is prohibited.

11.2.2. Using brush or fiber rubbing FL fabric for removing stains is prohibited.

11.2.3 Using ball-pen for marking panels is prohibited.

12. Garment Test

All styles must provide below test reports with passed result before release shipment. 1pc finished garment per- style is required.

12.1. Suter test : All critical points passed water test 1.5 psi/ 1min

12.2. Wash test: Wash the units been passed water test, 3 cycles wash is required.

Garment Test to conduct	Standard Test Method	Requirement	Development	Bulk
Functional claims testing (according to MPS) including,			-	**Fabric mill submit to 3rd party for testing**
- Waterproofness (Low Range and High Range Hydrostatic Pressure) before and after wash 20 cycles	*AATCC 127 Option 2 ASTM D751*	*3000 mm / 1500 mm minimum 25 psi / 15 psi*		
- Air Permeability	*ASTM D737*	*0.5 cfm minimum*		
- Breathability (Moisture Vapor Transmission Rates)	*ASTM E96 B (Initial only)*	*800 g/m²/24 hours minimum*		
- Durable Water Repellency (Spray Test)	*AATCC22*	*Initial > 90 after 20*wash > 70*		
UL Verification Testing			X	(X)
Standard test method for Liquid Penetration Resistance of Protective Clothing under a shower spray while on Manikin in 60 min rain cycle				

Full Garment test* including,				
- Dimensional Stability to Washing	AATCC 135	-3.0 % to +3.0%	X	X
- Distortion after Washing	AATCC 179 Option 1	3.0% maximum		
- Appearance after Washing	VFOI-004 (after 3 Washes)	Class 4.0		
- Seam Properties	ASTM D1683 (Garment)	Refer to FL MPS		
- Attachment Strength of Applied Trims	ASTM F963	20lbf for 10 secs		
- Colorfastness test	AATCC 107	Self Staining Requirement: Class 5.0 as applicable. Refer to FL MPS		
- pH value	AATCC 81	Colors 5.0-8.0; White 4.0-7.0		
Pull test on stress point of particular construction	ASTMD7506/D7506M-09	15 lbs	(X)	(X)
Appearance after oven storage 70C within +/-2C, Relative humidity 65%+/-2% in 24 hrs	VFOI-007	Class 4	-	X

X – Apply the test,
(X) – Conditionally apply the test
** - Full garment test requirement can be found in the TNF Quality Testing Manual SOP section 3.3.*

13. Final Inspection

13.1. AQL sampling plan: All FL styles must undergo final inspection follow by AQL 2.5 Tightened sampling plan.

AQL SAMPLING (MIL-STD-105 LEVEL II)						
Lot Size Range		NORMAL SAMPLE		TIGHTENED SAMPLE		ZERO DEFECT (C=0)
MIN	MAX	Sample	ACC#	Sample	ACC#	
2	8	5	0	8	0	5
9	15	5	0	8	0	5
16	25	5	0	8	0	5
26	50	5	0	8	0	7
51	90	20	1	32	1	11
91	150	20	1	32	1	11
151	280	32	2	32	1	13
281	500	50	3	50	2	16
501	1200	80	5	80	3	19
1201	3200	125	7	125	5	23
3201	10000	200	10	200	8	29
10001	35000	315	14	315	12	35
35001	100000	500	21	500	18	40

NOTES:

1. Selected sample must correspond to the applicable Lot Size Range.
2. Default sample is "Normal" "Tightened" sampling is employed to address repeated non-compliance issues.
3. Zero Defect sampling may be utilized for typical "Go/No-Go" product characteristics. The ACC# for Zero Defect sample is always zero. Lots failing Zero Defect sample audit may require completing full

AQL Normal Sample audit in order to verify defect quantity is greater than the listed Normal AQL ACC#

13 Packing

The MCQ will be adjusted based on style, recommend carton size table as below:

Carton type	Destination	Carton Dimension
IR7	USA / CAN	Full MCQ carton: 28" x 15" x 11.5" Less MCQ carton: 28" x 15" x 5"
Out of carton D	EU	Full MCQ carton: 70cm x 38cm x 29cm Less MCQ carton: 70cm x 38cm x 15cm



Thank You

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